

Vayu VY-6

Human Factors Description

VAY-HFD-0001 - Issue 1

Single-pilot, six-passenger electric vertical take-off and landing air taxi. Demonstration article: the Vayu VY-6 is a representative aircraft constructed by Safety Lab Aero and is not the design data of any manufacturer. This document is a companion to VAY-SDD-0001 and describes the crew, the flight deck and the task, where that document describes the equipment.

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Introduction

This document describes the human factors context of the Vayu VY-6, a six-passenger electric vertical take-off and landing air taxi intended for short urban legs between city vertiports. It sets out the flight deck, the tasks the pilot performs across a flight, the way the aircraft presents information and alerts, and the interface passengers experience during boarding, flight and disembarkation. It is written to support downstream analysis of how tasks and information are arranged between the pilot and the aircraft's systems; it does not itself state conclusions about that arrangement.

The VY-6 is flown by one pilot occupying the forward centerline seat, with up to six passengers seated in the cabin behind. There is no cabin crew member on board. The aircraft operates at a high daily tempo, flying eight to fourteen legs a day of twenty to thirty kilometres each, by day and by night, with turnarounds at the pad of about ten minutes during which passengers board and leave while the aircraft remains close to buildings, other pads and members of the public.

The material that follows covers the operating environment, the physical and display layout of the flight deck, the division of activity between the pilot and the aircraft's automation as it is actually carried out in flight, the pilot's tasks phase by phase from pre-flight to shutdown, the philosophy behind how the aircraft draws the pilot's attention to conditions that need it, the passenger-facing interface, and the training route by which a pilot becomes qualified to operate the type. An appendix lists named crew tasks against the phase in which each occurs, with a short description of what the task involves.

1 Scope and Crew Concept

This document covers the VY-6 as configured for revenue passenger service on urban air taxi routes: a lift-plus-cruise electric vertical take-off and landing aircraft with a maximum take-off mass of 2,600 kg and a payload of 680 kg, seating one pilot and six passengers. The aircraft has eight lift rotors carried on two booms for hover and two wing-mounted propellers for forward flight, and transitions between the two lift regimes once in each direction on every flight. It is a representative demonstration article prepared to support analysis of the human factors questions raised by a small, high-tempo, single-pilot air taxi, and does not describe a certificated product.

The pilot occupies the forward centerline seat and is the only crew member on board. The cabin behind holds up to six passengers in a two-two-two arrangement, with no dedicated cabin attendant present on any flight. Passengers board and leave through cabin doors on the pad under the pilot's observation, and the pilot is responsible for confirming the cabin is ready before departure and for managing anything that arises in the cabin during flight, in addition to flying the aircraft. This document describes the tasks that fall to the pilot under this concept, the tasks the aircraft's systems carry out on their own, and the tasks where pilot input and system behavior are interwoven, without characterizing how the resulting arrangement should be judged.

The VY-6 is flown under fly-by-wire control through a centre inceptor that commands flight path and a left-hand lever that commands energy, backed by flight control computers, actuation electronics and dual data buses, with no mechanical or hydraulic path retained as a fallback if the electronic path is lost. Two large display surfaces in front of the pilot and a smaller standby display carry the information the pilot uses to fly, navigate, manage aircraft systems and monitor the cabin. The aircraft is intended to be flown from a single flight deck seat on every leg of a duty day that may run to the low teens of departures, each preceded by a short turnaround on a pad shared with other traffic and with the public nearby.

The scope of this document extends from the pilot's arrival at the aircraft before a duty day, through each flight leg and turnaround, to the end of the last leg of the day, together with the conditions that depart from the normal sequence: system faults, environmental conditions outside the ordinary range, and situations calling for use of the aircraft's emergency recovery system. It addresses the flight deck as installed, the passenger cabin so far as the pilot interacts with it, and the vertiport environment so far as it shapes what the pilot must do and notice. It does not address aircraft structural design, propulsion system internals, or air traffic

procedures except where the pilot's own actions and observations are the subject.

2 Operating Environment

The VY-6 operates from elevated pads built into the urban fabric: rooftop structures, purpose-built towers and repurposed platforms, most of them small relative to the aircraft's footprint and surrounded closely by buildings of varying height. The immediate airspace around a pad is frequently disturbed by the buildings themselves, producing gusts, shear layers and recirculating flow that can change from one approach to the next depending on wind direction and the traffic passing nearby. A pad that is calm on one leg can present a noticeably different air mass an hour later as the ambient wind shifts against the surrounding structures.

Vertiports sit inside dense urban airspace shared with other air taxis, helicopters and, at some sites, small fixed-wing traffic, all operating in proximity to tall structures, cranes, rooftop equipment and overhead wires. The pilot must maintain awareness of nearby traffic and obstacles throughout arrival, departure and the hover segments close to the pad, aided by the aircraft's detect-and-avoid system but without relying on that system as the sole means of separation. Radio and data-link communication with vertiport ground control and with other traffic is a continuous background task from the approach into a pad through to the departure from the next one.

The pad environment is populated. Passengers, ground handlers and members of the public may be present close to the aircraft while rotors are turning, on a platform that offers little standoff distance and often has other aircraft queued for the same pad. The pilot conducts arrivals and departures with this proximity constantly in view, watching for people or objects near the aircraft as much as for the aircraft's own flight path. Lighting on the pad ranges from full daylight to unlit night operation relieved only by the vertiport's own fixtures and the aircraft's landing lights, and the same pad can present very different visual cues by day and by night.

Weather relevant to the VY-6's operation includes gusting surface wind, precipitation that reduces visibility on approach, and temperature extremes that affect battery performance and available energy margins. Because legs are short and turnarounds brief, the pilot repeatedly re-evaluates conditions at the next pad while still completing tasks associated with the current one, with weather information refreshed frequently rather than reviewed once before a long cruise. Urban microclimates mean conditions can differ meaningfully between two pads only a few kilometres apart, so a benign departure does not guarantee a benign arrival on the same short flight.

Noise and community sensitivity shape where and how the aircraft is flown, with published arrival and departure paths intended to keep the aircraft over specific corridors rather than directly above occupied buildings wherever the pad layout allows it. The pilot follows these paths using the navigation display while also watching for traffic and obstacles that are not on any published path. Across a duty day of many legs, the pilot moves repeatedly between busy, close-in airspace at each end of a flight and a comparatively simple cruise segment in between, with the character of the operating environment changing on a timescale of minutes rather than hours.

3 Flight Deck Description

Flight Deck Arrangement

Panel and seating as described in section 3. Dimensions are not to scale.

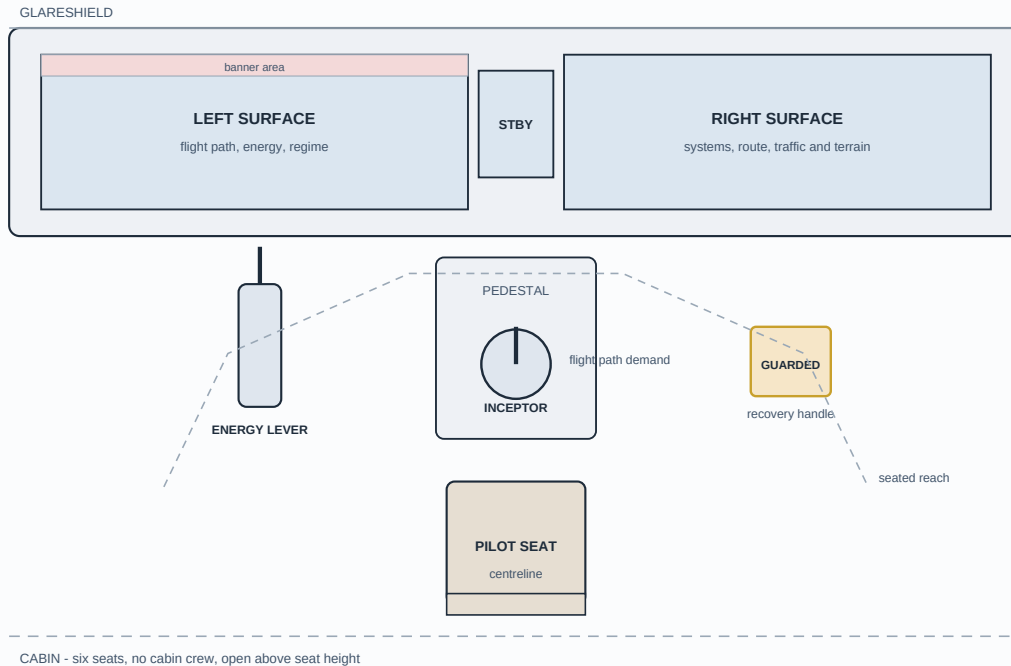


Figure 1 Flight deck arrangement. The panel, the inceptors and the seated reach envelope described in this section.

The flight deck seats the pilot on the aircraft centerline, facing forward, with the passenger cabin immediately behind. Two large display surfaces are mounted side by side ahead of the pilot, angled slightly toward the centerline, with a smaller standby display set below and between them. The two large surfaces are configurable: either can show flight path information, systems information, navigation and traffic information, or checklist and procedure content, and the pilot selects what appears on each through touch controls on the display bezels and through buttons on the inceptor and lever.

Primary flight information, including attitude, airspeed, altitude, vertical speed and flight path guidance, is normally shown on the display directly ahead of the pilot's primary line of sight, with engine and propulsion parameters, battery state, and system status pages available on the adjacent surface. A synoptic view of the aircraft's electrical, propulsion and thermal systems can be called up on either surface, showing the state of the battery packs, the lift rotors, the cruise propellers and the distribution system in a single schematic. The standby display presents a reduced set of attitude, airspeed and altitude information directly in front of the pilot, independent of the two main surfaces, for use if either or both of them become unavailable.

Aircraft control is through two hand controls. A centre inceptor, held with the right hand, commands flight path: pitch and roll rate in wing-borne flight, and translational rate and heading in hover, with the flight control computers translating inceptor position into the commands sent to the actuation system and the propulsion electronics. A left-hand lever commands energy: thrust and climb or descent rate in cruise, and vertical rate in hover, again interpreted by the flight control computers rather than acting on any mechanical linkage. Neither control connects to the aircraft's surfaces or motors by a direct mechanical path; both act only through the flight control computers, and there is no mechanical or hydraulic fallback if that electronic path is unavailable.

Around the inceptor and lever are buttons and switches for selecting flight modes, arming the transition between hover and wing-borne flight, and calling up checklists on the displays. A radio and data-link panel sits within reach for communication with vertiport control and other traffic, and a separate panel controls cabin lighting, door status indication and the passenger address function, giving the pilot a way to address the cabin without leaving the flight deck. An emergency recovery handle, distinct in shape and location from any other control, is positioned within reach of the pilot's seat and, when pulled, deploys the aircraft's ballistic

parachute; it is guarded to reduce the chance of inadvertent operation.

Door and cabin status, including whether each passenger door is fully closed and latched, is repeated on the flight deck displays so the pilot can confirm cabin readiness without a physical check of each door from the seat. A forward-facing camera view of the pad area and a rearward-facing view into the cabin are available on the displays, giving the pilot a way to see the area immediately around the aircraft and the state of the cabin without turning around or leaning to a window. Seat, restraint and general flight deck ergonomics are arranged so the pilot can reach every control needed for a flight, including the emergency recovery handle, without releasing the primary hand controls for more than a brief interval.

4 Automation and Pilot Roles

Once the pilot commands a flight path or energy setting through the inceptor or lever, the flight control computers translate that command into the individual signals sent to the actuation system, the lift rotors and the cruise propellers, continuously adjusting those signals to hold the commanded path and speed against gusts, air density changes and the aircraft's changing mass as batteries discharge. The pilot does not set individual rotor speeds or control surface positions directly at any point in a flight; those are computed and commanded by the flight control computers from the pilot's higher-level inputs on the inceptor and lever.

Stabilization in hover, including the constant small corrections needed to hold a position against gusts and shear near buildings, is carried out by the flight control computers without pilot action beyond holding the inceptor at the desired position; the pilot's inputs express an intended movement or hold, and the computers produce the changing rotor commands that keep the aircraft where the pilot has asked it to be. In wing-borne cruise, the flight control computers similarly hold the commanded speed and flight path, adjusting propeller thrust and control surfaces continuously in response to turbulence and to the small trim changes produced as energy is consumed and the aircraft's mass and centre of gravity shift over a leg.

The transition between hover and wing-borne flight is sequenced by the flight control computers once the pilot arms and commands the transition: rotor thrust is reduced and propeller thrust increased, or the reverse, along a profile calculated from airspeed and other flight parameters, with the lift rotors brought to a stop and their blades locked in position for cruise, or released and spun up again ahead of hover, at points in the sequence chosen by the flight control computers rather than by pilot timing. The pilot retains the ability to pause or reverse a transition through the inceptor and lever if conditions call for it, and monitors the transition's progress on the primary flight display, but does not command the individual steps of the sequence.

Navigation along the route between vertiports, including the lateral path and, where a profile is loaded, the vertical path, can be followed automatically once the pilot selects and activates a route, with the flight control computers steering the commanded path and the pilot free to intervene through the inceptor at any point, which suspends the automatic path following until re-engaged. The pilot chooses the route to fly, decides when to follow it automatically and when to hand-fly a segment, particularly during the visually demanding final approach into a vertiport, and makes the judgment calls about traffic, weather and pad conditions that lie outside what the automatic path following is set up to evaluate.

Detection of nearby traffic and obstacles is performed continuously by the detect-and-avoid system, which processes sensor returns and presents traffic on the navigation display together with an indication of anything assessed as requiring the pilot's attention; the system does not manoeuvre the aircraft on its own, and the decision to alter course, hold position or continue remains the pilot's, informed by what the system shows and by the pilot's own view outside the aircraft. Throughout every phase, the pilot retains a continuously available means to take direct control of flight path and energy through the inceptor and lever, overriding whatever automatic function was previously engaged, and the aircraft is designed so that this override is available without a mode change that itself requires more than a brief action to select.

5 Crew Tasks by Phase

Phases of a Sector

The order in which section 5 describes the crew tasks.

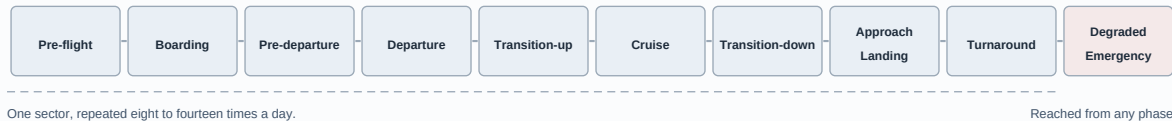


Figure 2 Phases of a sector, in the order this section describes them.

Pre-flight and power-up begins when the pilot arrives at the aircraft, walks around it to look for visible damage, leaks, loose panels or anything abnormal about the airframe, booms, rotors and propellers, and checks that the cabin, doors and emergency equipment are in the expected condition. The pilot then powers up the flight deck, watching the displays as the flight control computers, navigation and air data systems, communication equipment and the propulsion and battery systems initialize, and works through a series of checks presented step by step on the display, confirming battery state of charge, system status indications and control response before the aircraft is considered ready for passengers. Weather and route information for the upcoming leg is reviewed during this period, drawing on data-link updates that may have changed since the previous leg.

Passenger boarding follows, with the pilot typically remaining in the flight deck seat while passengers approach and are directed to their seats, watching the cabin camera view and listening for anything unusual, and using the passenger address function to give any safety information the operation calls for. The pilot confirms each door shows closed and latched on the flight deck display, cross-checking with the cabin camera view where doubt exists, and confirms passenger count and loading are within what the flight was planned for. Because the pad is shared with other traffic and passengers waiting nearby, the pilot also keeps half an eye on activity outside the aircraft throughout boarding.

Pre-departure checks bring together the power-up sequence, the boarding confirmation, and a final review of weather, traffic and pad conditions before requesting clearance to depart. The pilot works through a final checklist covering flight control response, navigation readiness, communication with vertiport control, and confirmation that the route and any loaded vertical profile match the intended departure. Once clearance is received, the pilot arms the aircraft for vertical departure and gives passengers, through the address function, whatever indication the operation calls for.

Vertical departure has the pilot lifting the aircraft from the pad using the lever to command a climb rate and the inceptor to hold position, watching the primary display for height, rate of climb and any drift, and watching outside, aided by the forward camera view, for obstacles, traffic or anything on the pad that changes as the aircraft lifts clear. Wind and building-induced gusts are most noticeable in this phase, close to structures, and the pilot makes the corrections needed to keep the aircraft clear of the pad edge and nearby obstacles while climbing to the height at which transition will begin.

Transition to wing-borne flight is armed and commanded by the pilot once clear of the pad environment and at the height and speed called for by the procedure, after which the flight control computers carry out the sequence of reducing lift rotor thrust and increasing cruise propeller thrust while the pilot monitors airspeed, height and the transition's progress on the primary display, ready to pause or reverse it through the inceptor and lever if the aircraft's behavior or the surrounding air does not match expectations. The pilot also continues watching for traffic, since the aircraft is climbing and accelerating through shared airspace.

Cruise is the most settled phase of the flight, with the pilot following the loaded route, monitoring battery state, systems status and traffic on the displays, communicating with vertiport control and other traffic, and periodically reviewing conditions expected at the destination pad. Because legs are short, cruise on the VY-6 is brief relative to the phases either side of it, and the pilot uses much of this period to prepare for the

approach and turnaround to come.

Transition back to hover is commanded by the pilot approaching the destination vertiport, again carried out as a sequence by the flight control computers once armed, with lift rotors spun up as propeller thrust is reduced, while the pilot monitors airspeed, height and the sequence's progress and watches ahead for the vertiport, its pad markings and any traffic in the approach path. This transition typically occurs some distance from the pad, so the pilot manages the aircraft's changing flight characteristics while beginning to assess the approach to follow.

Approach and landing on the pad calls for the pilot to fly the aircraft down to a small, often partially obstructed landing area, watching wind indications, pad markings, and any people or equipment on or near it, while managing building-induced turbulence and shear that are frequently most pronounced close to the surrounding structures. The pilot uses the inceptor to hold position and correct drift and the lever to control descent rate, cross-checking height and rate of descent on the display against the visual picture outside, and is prepared to break off and go around if the pad is not clear or the approach does not develop as expected. Touchdown is confirmed both by the aircraft's own indications and the pilot's own sense of the landing, after which the pilot brings rotor thrust to idle and completes the checks needed to leave the aircraft safe on the pad.

Turnaround compresses a number of tasks into a short period on the pad: confirming the aircraft is safe for ground personnel and passengers to approach, opening or confirming open doors for disembarkation, watching passengers leave through the cabin camera view and windows, attending to anything left behind or any assistance a passenger needs, reviewing battery and system status to judge readiness for the next leg, and beginning the boarding sequence for the next passengers, all while other aircraft may be waiting for the same pad and ground personnel move around the aircraft. The pilot manages this without another crew member, drawing on the display's step-by-step checks and the cabin and pad camera views to track the surroundings while working through them.

Degraded or emergency operation covers situations that depart from the normal sequence: a system fault reported on the displays, wind conditions exceeding what a normal approach can accommodate, a loss of communication with vertiport control, or, rarely, a situation serious enough to warrant the ballistic parachute. In each case the pilot's task begins with recognizing the condition from what the displays, controls and the aircraft's behavior are showing, working through the guidance the display presents, and deciding on a course of action that may include diverting to another pad, holding until conditions improve, continuing with reduced capability, or pulling the emergency recovery handle. Communication with vertiport control and, where relevant, with passengers over the address function continues alongside these actions, all carried out without another crew member to share the assessment or the flying with.

6 Alerting Philosophy

The VY-6 draws the pilot's attention to a condition through a combination of what appears on the displays and what is heard in the flight deck, arranged so that the most pressing conditions are the hardest to miss and the least pressing are the easiest to set aside without distraction. A condition requiring the pilot to act without delay is presented with a banner across the top of the primary display in a colour reserved for that purpose, accompanied by a repeating tone that continues until the condition is addressed or clears, and, for a small number of conditions bearing on immediate control of the aircraft, a change in the feel of the inceptor or lever that the pilot notices through the hand controls themselves. A condition that needs attention but does not call for immediate action is shown in a different colour on the same banner area, generally accompanied by a single tone rather than a repeating one, drawing the eye without demanding it be held there.

Information that the pilot may want to know about, but that does not call for a change in what the pilot is doing, is presented as text within the relevant system page rather than on the attention-getting banner, in a colour distinct from either of the two categories above, and without any accompanying sound. The intent is that a glance at the primary display tells the pilot whether anything currently needs a response, without the

pilot having to search through system pages to find out, and that the sound and feel cues reserved for the most pressing conditions are not spent on matters that can wait for a convenient moment to review.

Alerts are grouped and, where more than one condition is active at once, ordered on the display so the pilot can see at a glance which condition is newest or most directly relevant to what the aircraft is currently doing, rather than being presented as an undifferentiated list that the pilot must read in full to find the item that matters most. Where a fault affects a system the pilot might otherwise try to use, the display also shows what remains available, so the pilot is not left inferring the aircraft's remaining capability from the alert text alone. Checklist guidance associated with a given condition can be called up directly from the alert itself, placing the relevant steps on the adjacent display surface without the pilot having to search for the right procedure.

Because the VY-6 flies close to the ground and to structures for a significant part of every leg, alerts bearing on height, rate of descent, proximity to terrain or obstacles, and nearby traffic are presented promptly and are designed to be noticed even while the pilot's visual attention is largely outside the aircraft during an approach or departure; the tones used for these conditions are chosen to be distinguishable from other tones used elsewhere in the flight deck so the pilot can tell, without looking, roughly what kind of condition has just been raised. Spoken callouts are used for a limited set of conditions where a short phrase communicates the situation more directly than a tone alone, such as an automatic call of remaining height during the final part of an approach.

The pilot can silence a repeating tone once a condition has been seen, without clearing the underlying banner, so that a persistent noise does not compete with other tasks once the pilot is already aware of the situation and is working through it; the visual indication of the condition remains until it is actually resolved. During the turnaround on a pad, a period in which the pilot is dividing attention between the cabin, the pad environment and preparation for the next departure, the aircraft continues to raise conditions through the same set of cues used in flight, and nothing about the turnaround period changes how a given condition is presented, so a fault arising while doors are open and passengers are moving is shown to the pilot in the same way it would be in any other phase.

7 Passenger Interface

Passengers approach the VY-6 on the pad and board through cabin doors under the pilot's observation, with no cabin attendant to guide them, so the interface presented to passengers is built into the cabin itself and into the address function the pilot operates from the flight deck. Seats face forward in a two-two-two arrangement behind the pilot, each with its own restraint, and placards at each seat show how the restraint is fastened and where the nearest exit is, in the same way a passenger would expect from any small aircraft cabin. A short recorded or live safety briefing, delivered through the cabin speakers and triggered by the pilot, covers restraint use, exit location, and the location of the aircraft's brace position guidance before every departure.

Cabin lighting, including a change in lighting used to draw attention during an unusual situation, and door status indication are visible to passengers as simple, unambiguous states rather than as technical information, so a passenger can tell whether a door is open or closed and whether the cabin is in a normal or an unusual condition without needing to interpret anything. A small display or indicator visible from the passenger seats may show flight progress information such as remaining time or distance to the next vertiport, giving passengers a sense of where they are in the flight without requiring any interaction with the aircraft's own systems.

The pilot addresses the cabin directly through the passenger address function at points across a flight: before departure, during any unusual condition that passengers should be aware of, and before arrival, and can hear sound from the cabin through the same system, giving a route for a passenger to attract the pilot's attention if needed, for example by speaking toward a microphone location identified in the cabin. The rearward-facing cabin camera view on the flight deck displays gives the pilot a way to check the cabin's state, including whether passengers are seated and restrained, without leaving the flight deck seat, and this view is used

routinely before departure and during any part of a flight where the pilot has reason to check on the cabin.

Because turnarounds are short and passengers board and leave close to a pad shared with other aircraft, the passenger interface is also concerned with keeping departing and arriving passengers away from rotors, propellers and other hazards on the pad; door design, marked paths from the cabin to a waiting area, and ground personnel at larger vertiports all contribute to this, with the pilot's own observation of the pad through the forward camera and cabin doors serving as a further check that the area passengers must cross is clear before doors are opened. Passengers are not expected to operate any aircraft system themselves beyond fastening their own restraint and, where fitted, opening a door under guidance once the aircraft indicates it is safe to do so.

For passengers with reduced mobility or particular assistance needs, the cabin layout and door arrangement are intended to allow boarding and disembarkation with help from ground personnel or the pilot without requiring the aircraft to be reconfigured, and the pilot's address function and cabin camera view give a way to confirm such a passenger is settled and secure before departure. Across a day of many short legs, the passenger interface is designed to convey the small amount of information a passenger actually needs quickly and plainly, on the expectation that most passengers will experience the aircraft for only the few minutes of a single urban leg and will not have had the flight explained to them beyond the briefing given before each departure.

8 Training and Qualification

A pilot destined for the VY-6 begins with the qualifications and experience that let them fly a fly-by-wire aircraft with no mechanical reversion, and moves through a course that covers the aircraft's systems, its two flight regimes and the transition between them, its displays and controls, and the procedures used across a normal flight and the range of degraded and emergency situations described elsewhere in this document. Classroom and computer-based instruction covers how the flight control computers, propulsion, energy storage, navigation and communication systems work and fail, how the displays present information in each case, and how the inceptor, lever and other controls behave in hover, in cruise and during transition.

A significant part of training uses a full-motion or fixed-base simulator representing the VY-6 flight deck, in which the trainee flies complete legs including departure, transition, cruise, transition back to hover, approach and landing, and turnaround, repeated under a range of wind, visibility and traffic conditions representative of the urban vertiport environment. The simulator is also used to present system faults and degraded conditions the trainee would rarely or never encounter in the aircraft itself, letting the trainee practice recognizing a condition from the displays and controls, working through the guidance presented, and deciding on and carrying out a course of action, including use of the emergency recovery system, without exposing the aircraft or the trainee to genuine risk.

Because the pilot operates without another crew member, training gives particular attention to managing a full flight from departure to turnaround alone: dividing attention between flying the aircraft, communicating with vertiport control and other traffic, watching for traffic and obstacles, and attending to the cabin and passengers, all without another person to share any of these tasks or to notice something the pilot has missed. Training scenarios include busy turnarounds with passengers boarding while other aircraft are queued for the same pad, approaches into pads affected by building-induced turbulence, and the arrival of a system condition at an inconvenient point in a flight, so the trainee builds familiarity with the kind of situation described in the crew tasks and alerting sections of this document before encountering it during revenue operation.

Once initial training is complete, a pilot demonstrates the required knowledge and skill to a check pilot, covering normal operation of the aircraft across its flight regimes and the ability to recognize and respond appropriately to a representative set of degraded and emergency conditions, before being released to fly passengers. Recurrent training at defined intervals thereafter revisits both normal procedures and infrequently encountered situations, including transition anomalies, loss of a display surface, and use of the

emergency recovery system, so that skills exercised rarely in daily line flying are refreshed periodically in the simulator rather than left to decay between the rare occasions they might actually be needed.

Route and vertiport familiarity is treated separately from aircraft type training: a pilot new to a particular city or a particular vertiport receives briefing material and, where practical, an introduction to the specific pads, approach and departure paths, and local traffic patterns involved, recognizing that the building-induced turbulence, pad size, and visual references at one vertiport can differ substantially from another even though the aircraft and its systems are unchanged. Ongoing qualification also depends on maintaining currency through the high tempo of daily operation itself, with the frequent repetition of departures, transitions, approaches and turnarounds across many short legs a day serving, alongside formal recurrent training, to keep the pilot practiced in the tasks this document describes.

Appendix A - Task Index

Every crew task named in section 5, listed flat and tagged with the phase in which it is performed. The list is an index to the narrative rather than a procedure: it says what the pilot does and when, and it does not say how long any of it takes or how the tasks divide between the pilot and the aircraft.

Task	Phase	Description
Exterior walkaround	Pre-flight	Visual inspection of airframe, booms, rotors, propellers and cabin exterior for damage, leaks or loose panels before power-up.
Flight deck power-up	Pre-flight	Energizing the flight deck and observing initialization of flight control, navigation, communication and propulsion systems.
System status review	Pre-flight	Working through displayed checks of battery state, control surface and rotor response, and overall system status.
Weather and route review	Pre-flight	Reviewing current weather, route and any updated data-link information for the upcoming leg.
Cabin readiness check	Pre-flight	Confirming cabin, doors and emergency equipment are in the expected condition ahead of boarding.
Passenger monitoring during boarding	Boarding	Observing passengers approaching and settling in the cabin using the cabin camera view and direct awareness.
Safety briefing delivery	Boarding	Triggering or delivering the recorded or live safety briefing over the cabin address function.
Door status confirmation	Boarding	Confirming each cabin door shows closed and latched on the flight deck display before proceeding.
Passenger count and load check	Boarding	Confirming passenger count and loading are within the limits planned for the flight.
Pad environment scan	Boarding	Watching activity on and around the pad, including other aircraft and ground personnel, during boarding.
Final systems checklist	Pre-departure	Working through the displayed pre-departure checklist covering controls, navigation and communication readiness.
Route and profile confirmation	Pre-departure	Confirming the loaded route and any vertical profile match the intended departure.
Departure clearance coordination	Pre-departure	Communicating with vertiport control to obtain and confirm clearance to depart.
Departure announcement	Pre-departure	Advising passengers over the address function that departure is about to begin.
Vertical lift-off control	Departure	Commanding climb rate and holding position over the pad using the lever and inceptor during initial lift-off.
Obstacle and pad clearance watch	Departure	Watching outside the aircraft and the forward camera view for obstacles or activity near the pad while climbing.
Gust and shear correction	Departure	Making manual corrections for building-induced gusts and shear encountered close to structures during climb-out.
Traffic awareness during climb	Departure	Monitoring nearby traffic on the navigation display and visually while climbing clear of the vertiport.
Transition arming and command	Transition-up	Arming and commanding the transition from hover to wing-borne flight once height and speed criteria are met.
Transition progress monitoring	Transition-up	Watching airspeed, height and the transition sequence on the primary display as it proceeds.

Task	Phase	Description
Transition pause or reversal readiness	Transition-up	Being ready to pause or reverse the transition sequence through the inceptor and lever if conditions warrant.
Route following management	Cruise	Selecting and monitoring automatic route following, or hand-flying, along the leg to the destination.
Battery and systems monitoring	Cruise	Periodically reviewing battery state and systems status displays during the cruise segment.
Traffic and communication management	Cruise	Communicating with other traffic and vertiport control and monitoring traffic displays through the leg.
Destination conditions review	Cruise	Reviewing weather and expected conditions at the destination pad in preparation for arrival.
Transition arming and command (arrival)	Transition-down	Arming and commanding the transition from wing-borne flight back to hover on approach to the destination.
Approach picture assessment	Transition-down	Beginning visual assessment of the destination vertiport and approach path while the transition proceeds.
Pad and wind assessment	Approach/Landing	Evaluating wind indications, pad markings and clearance of the landing area during approach.
Position and descent control	Approach/Landing	Using the inceptor and lever to hold position, correct drift and control descent rate to touchdown.
Go-around decision	Approach/Landing	Deciding to break off and go around if the pad is not clear or the approach does not develop as expected.
Touchdown confirmation and shutdown to idle	Approach/Landing	Confirming touchdown and bringing rotor thrust to idle once the aircraft is settled on the pad.
Post-landing safety confirmation	Turnaround	Confirming the aircraft is in a safe state for ground personnel and passengers to approach.
Disembarkation monitoring	Turnaround	Watching passengers leave the cabin using the camera view and direct observation.
Turnaround systems review	Turnaround	Reviewing battery and system status to judge readiness for the next leg.
Next-leg boarding restart	Turnaround	Beginning the boarding and pre-departure sequence for the next set of passengers.
Pad traffic coordination	Turnaround	Coordinating with other aircraft and ground personnel waiting for or using the same pad.
Fault recognition and checklist response	Degraded/Emergency	Recognizing a reported system condition and working through the guidance presented on the display.
Diversion or hold decision	Degraded/Emergency	Deciding whether to divert to another pad, hold away from a pad, or continue with reduced capability.
Passenger notification during an unusual condition	Degraded/Emergency	Communicating relevant information to passengers over the address function during a degraded condition.
Emergency recovery system deployment	Degraded/Emergency	Assessing the situation and, where warranted, pulling the guarded emergency recovery handle.